

Anteater Poker User Manual

Version 0.1 Alpha Draft

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Glossary

- **All-In** – A player bets all their remaining chips on a single hand. They cannot be forced out of a pot because they run out of chips; instead, a side pot may be created.
- **Ante** – A small, mandatory bet required from all players before a hand begins to seed the pot.
- **Bet** – The act of placing an initial wager during a betting round.
- **Big Blind** – A mandatory bet posted by the player two seats to the left of the button. It is usually double the size of the small blind.
- **Bluff** – To bet or raise with a hand that is unlikely to be the best, with the intent of making opponents fold better hands.
- **Board (The Community Cards)** – The five cards dealt face-up in the center of the table that all players use in combination with their hole cards.
- **Burn Card** – The top card of the deck that is set aside face-down before dealing community cards to prevent cheating or card marking.
- **Call** – To match the current highest bet or raise in the betting round.
- **Check** – To pass the action to the next player without betting, permitted only if no one else has made a bet in the current round.
- **Community Cards** – Cards dealt face-up that are shared by all players to form their best five-card hand. (See: Flop, Turn, and River).
- **Flop** – The first three community cards dealt simultaneously after the first round of betting.
- **Fold** – To discard a hand and forfeit interest in the current pot.
- **Hole Cards** – The private cards dealt face-down to each player (two in Texas Hold'em).
- **Main Pot** – The primary pool of money that all active players are eligible to win.
- **Muck** – To fold a hand without showing the cards to the other players; also refers to the pile of discarded cards.
- **Pot** – The total amount of chips or money wagered by players during a single hand.
- **Raise** – To increase the size of the current bet, forcing subsequent players to match the new amount, fold, or raise again.
- **River** – The fifth and final community card dealt.

- **Showdown** – The final phase after all betting rounds are complete, where remaining players reveal their cards to determine the winner.
 - **Side Pot** – A separate pot created when a player goes "All-In," but other players continue betting. The All-In player is only eligible to win the pot they contributed to.
 - **Small Blind** – A mandatory bet posted by the player immediately to the left of the button.
 - **Turn** – The fourth community card dealt face-up.
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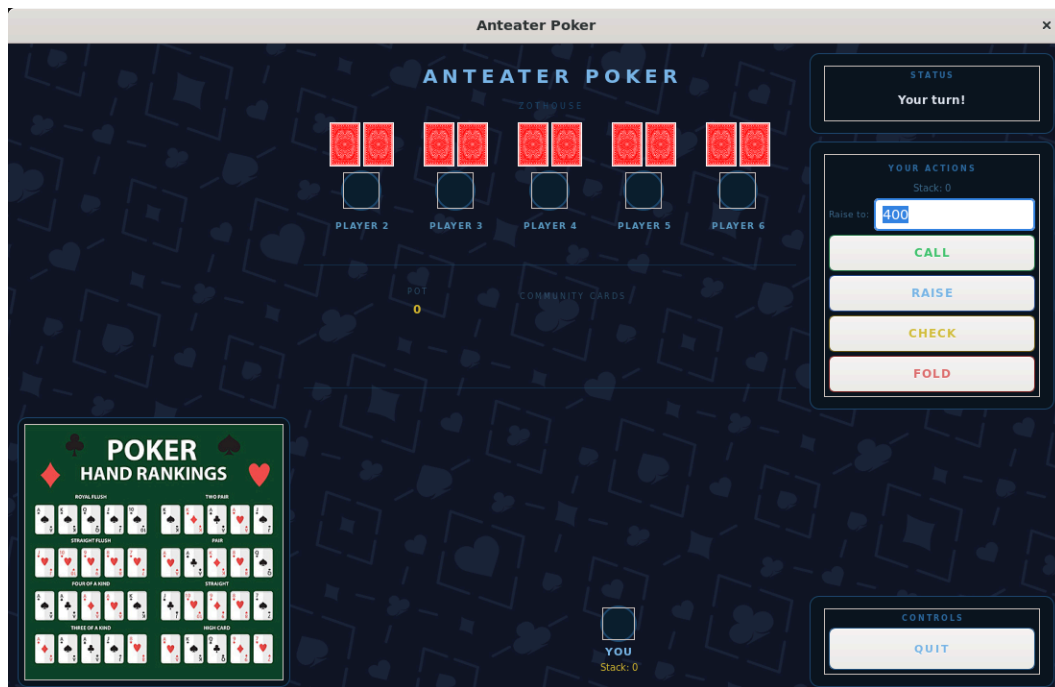
Hand Rankings (Highest to Lowest)

- **Royal Flush**: A, K, Q, J, 10 of the same suit.
- **Straight Flush**: Five consecutive cards of the same suit.
- **Four of a Kind (Quads)**: Four cards of the same rank.
- **Full House (Boat)**: Three of a kind combined with a pair.
- **Flush**: Five cards of the same suit, not in sequence.
- **Straight**: Five consecutive cards of different suits.
- **Three of a Kind (Trips/Set)**: Three cards of the same rank.
- **Two Pair**: Two different pairs in one hand.
- **One Pair**: Two cards of the same rank.
- **High Card**: The highest card in the hand when no other ranking is met.

1. Online Poker

Anteater Poker is an online multiplayer card game based on Texas Hold'em. A central server hosts the table and several graphical client programs connect across the network. Players log in with a display name, choose a seat, receive private cards, take turns, and compete for points.

1.1. Usage Scenario



Above is a possible screenshot of what a working online poker match would look like. The user's hole cards would be shown in front of them, with action buttons giving the user the choice to bet, call, or fold. The user will have the ability to see other players' bets and be able to make their choice based on that. The player is expected to run Zoom or another voice chat in parallel if the group wants live conversation.

1.2. Goals

- Allow users to play against each other online
- Develop a poker bot that players can play against
- Follow official poker rules with additional developer added rules.
- Provide a clear and interactive game experience.
- Provide a graphical interface on both the player side and the server control side.
- Create a program that can be built, tested, and submitted on the course Linux servers.

1.3. Features

Required Features

1. Allow all users to connect to server, submit their name, and choose their seat
2. Point tracking for all players
3. Program randomly shuffle and deal cards to players
4. Program needs to properly end game and award points to the winner
5. GUI for the player/client side and the server/control-room side.
6. Working poker bot that takes up at least one seat
7. GIT version control
8. TCP/IP communication via sockets to communicate between server and user

Optional / Planned Features

1. Consumable items that user can use for buffs or debuffs
2. Timer for each round
3. Scoreboard at the end of each round showing player rankings and what hand they had
4. Ability to choose which version of poker to be played such as five card draw, Omaha Hold 'em, or Seven Card Stud

2. Installation

2.1. System Requirements

- Operating System: Windows / macOS / Linux
- Compiler: GCC (C language), make, gtar, and standard C libraries.
- GUI Libraries: GTK 3
- Memory: At least 512 MB recommended.
- Storage: At least 50 MB available space.

2.2. Unpacking and configuration

Currently, the program has not yet been developed, so a complete setup and configuration process is not available. However, the expected installation procedure is as follows:

Already compiled:

1. Download and extract the compiled program files.
gtar xvzf Poker_V1.0.tar.gz
2. Open terminal and navigate to project directory.
cd poker
3. Skip to step 4 to launch the program.

Compiling from Source:

1. Download and extract the raw source files.
gtar xvzf Poker_V1.0_src.tar.gz
2. Open terminal and navigate to project directory.
cd poker
3. Compile the Source code.
make

4. Before starting a game, the host should choose a port and confirm that other players can reach the server. The default port will be 10010. If the port is blocked or already in use, choose another port and share it with the rest of the players.

Example Server Start:

./bin/poker_server --port 10010 --table "ZotHouse"

Example Client Start:

./bin/poker_client --host server name or ip --port 10010 --name Player

Configuration Options:

Option	Example	Purpose
--host	--host crystalcove.ics.uci.edu	Server name or IP address used by client.
--port	--port 10010	TCP port used by both servers and client.
--name	--name Hamza	Optional player display name. If not added, the login window will prompt the user.
--bots	--bots 1	Number of bot players that the server may add to empty seats.
--points	--points 1000	Starting points for each player.
--log	--log logs/game.log	Location of game log for debugging.

Developers using the source package can run `make test` to execute unit tests for major modules and system tests for client-server behavior.

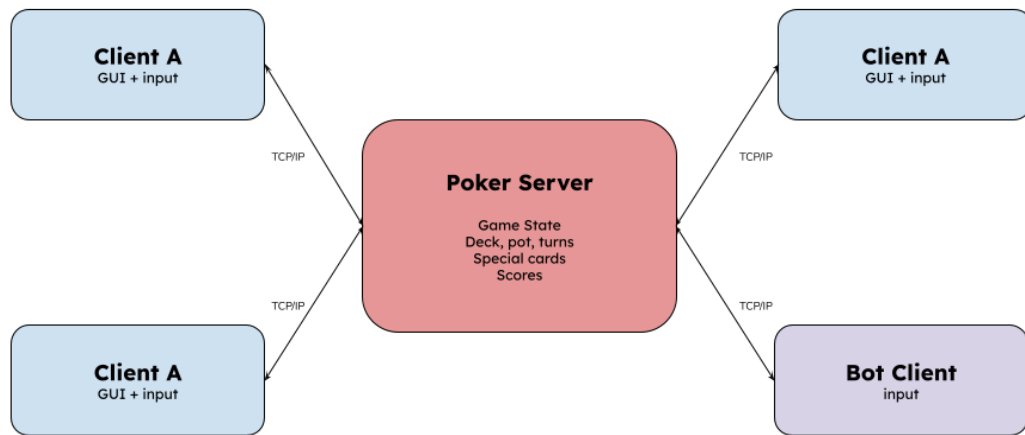
2.3. Uninstalling

- Delete the project folder
- Remove compiled files if needed.

3. Program Functions and Features

3.1. Client and Server Communication

Our implementation of Anteater Poker will use a central server, so clients never directly modify the game state. Instead, each client sends a requested action, and the server checks whether the action is legal. If it is legal, the server updates the game state and broadcasts the appropriate public information. Private cards are sent only to the matching player client.



Clients send actions. Server validates rules and broadcasts public game updates.

Figure 2. Program and Network communication between clients, bots, and the Server.

Message	Direction	Description
CONNECT	Client → Server	Request to open a table connection.
LOGIN	Client → Server	Send player display name.
SEAT_REQUEST	Client → Server	Request a specific open seat.
GAME_STATE	Server → Client	Public table information such as pot, scores, community cards, and whose turn it is.
PRIVATE_HAND	Server → Client	The two private cards for one player only.
ACTION_REQUEST	Server → Client	Requests an action from a given player for their turn.
PLAYER_ACTION	Client → Server	Player chooses a given action for their turn.
ERROR	Server → Client	Explains invalid input or illegal action.
SHOWDOWN	Server → Clients	Reveals hands and winner after the final betting round.

3.2. Dealer Choice and Card Distribution

At the start of a table, the server acts as the dealer. The server manages the game state, shuffles the deck, deals cards to players, reveals community cards, tracks turns, and awards points at the end of each round.

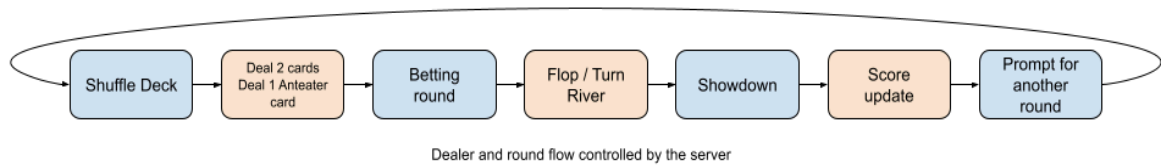


Figure 3. Dealer selection and card distribution flow.

Card distribution follows these rules:

1. The server creates a deck containing the standard 52 cards. A second Anteater deck of special cards is created.
2. The server shuffles the two decks using a random shuffle before each hand.
3. The server deals three private cards (2 normal, 1 anteater) to every seated active player or bot.
4. Private cards are transmitted only to the matching client.
5. The server manages betting after the private deal, after the flop, after the turn, and after the river.
6. The flop consists of three community cards. The turn and river each add one community card.
7. At showdown, the server evaluates all active hands and awards the pot to the winner.
8. The server prompts all players to choose whether they want to play another round or exit.

Expected user input: the user clicks the enabled action button and, for Raise, enters a point amount.

Expected program output: the client updates the table display, disables unavailable actions, and shows a status message such as “Waiting for Hamza” or “Your turn”

3.3. Poker Game Integration

The game uses traditional Texas Hold'em poker rules as the main structure. Each player begins with the number of starting points configured by the server. If no value is entered, the default starting amount is 1000 points. Players use their private cards together with the community cards to make the best possible five-card poker hand. The hand rankings from highest to lowest are shown in the glossary.

Anteater Rule: Special Ability Cards

At the beginning of each round, while the server deals two private cards to each player, the server may also assign an Anteater Ability Card to eligible or all players. Each ability card gives the player one special action that can be used during the round.

Poker Abilities:

All abilities are currently subject to change and may not appear in the final version of the game.

- **Sniff** – Because of the anteater's superior sense of smell, the player is able to "sniff out" one random private card from a chosen opponent.
- **Ant Trail** – Following an ant trail can lead to great things, the player may view either the suit or number of the next community card before it is revealed to anyone else.
- **Pose** – As a defensive action, the Anteater can stand in a protective pose to fend off predators, the player may use this card to cancel another player's special ability card played, regardless of whose turn it is.
- **Long Tongue** – Using the anteater's long tongue, you can swap one of your private cards with the top card of the deck.
- **Wild Grab** – Using the anteater's tail, this player may replace one of their private cards with a wildcard that can represent any card for the best possible poker hand.

Rules for Anteater Ability Cards:

- The server deals Anteater Ability Cards at the beginning of the round along with each player's private cards.
- Each Anteater Ability Card gives the player one special action.
- A player is only allowed to use one Anteater Ability Card per round.
- Once an ability is used, the card is marked as used and cannot be used again during that round.
- The server checks whether the ability is legal before applying it.
- Other players are notified that an Anteater ability was used, but private information stays hidden.
- Anteater Ability Cards do not count as poker cards and cannot be used to form a poker hand unless the ability calls for it.

3.4. Player Actions

The client will only enable actions that are legal for the current game state. The server remains the authority and rejects any illegal action sent by a modified or outdated client. The player still has specific actions that are considered legal during their turn or in some cases during other people's turn.

Action	When Available	Result
Check	No current bet must be matched.	The player stays in the hand without adding points.
Call	A current bet exists and the player has enough points to match it.	The player adds enough points to match the current bet.
Raise	A current bet can be increased and the player has enough points.	The player increases the bet by the selected amount.
Fold	Any betting turn.	The player leaves the current hand.
Use Ability	The player has an Anteater card available to use.	The selected Anteater Ability Card is activated if the server approves the action.
Leave	Any betting turn.	The player disconnects or becomes inactive.

The client should update or clear status messages after each step and turn.

3.5. Bots, Scoreboard, and End of Game

A bot player may be added by the server when seats are empty. The basic bot will be able to make legal decisions using simple rules, such as calling small bets, folding weak hands to larger raises, and checking when possible. The bot must use the same server rules as a human player. The bot will have the ability to use special Anteater cards as well and will do so accordingly.

Scoreboard behavior:

- Every player begins with the configured starting points.
- The pot is increased by calls and raises.
- Folded players cannot win the current pot.
- At showdown, the server awards the pot to the best hand.
- If all other players fold, the remaining player wins the pot immediately.
- At the end of a game session, the scoreboard ranks players by total points and may show each player's final hand from the last round.

The game can end when the host clicks End Game, when only one player remains with points, or when the team-configured maximum number of hands has been reached. The server should broadcast a final scoreboard before closing the table.

Back Matter

Copyright

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This software was developed as part of the EECS 22L course at the University of California, Irvine. It is intended for educational use only. Anteater Poker uses points only and must not be used for gambling.

Error Messages

Error Message	Meaning
Cannot connect to server	The client cannot reach the host or port.
Name already in use	Another connected player is using the same display name.
Seat unavailable	The selected seat is occupied or locked.
Not your turn	The client attempts an action while another player is active.
Invalid raise amount	Raise is below minimum, above available points, or not a number.
Illegal action	The requested action is not legal in the current state.
Server disconnected	The network connection closed unexpectedly.
Deck error	The server detected an invalid or exhausted deck state.

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